AI Trust Score for AntiCopyPaster

A confidence metric for AI-generated Extract Method refactorings

Definition

A weighted score that estimates how safe and reliable an AI-generated Extract Method refactoring is before the developer applies it.

Proposed Formula

AI Trust Score = 0.15·LLM + 0.20·Clone + 0.25·Usefulness + 0.20·Compilation + 0.15·Tests/Behavior + 0.05·Diff Safety

Why these percentages? V1 heuristic, not fixed: 25% Usefulness = core refactor quality; 20% Clone + 20% Compile = hard gates; 15% LLM + 15% Tests = supporting evidence; 5% Diff = reviewability check.

15% LLM Confidence

15%: useful signal, but model certainty ≠ code correctness

20% Clone Detection

20%: must first prove the pasted code is a real clone

25% Refactoring Usefulness

25%: highest weight because valid Extract Method is the main goal

20% Compilation Result

20%: broken code should block or sharply lower trust

15% Tests / Behavior

15%: verifies behavior, but test coverage may be incomplete

5% Diff Safety

5%: secondary reviewability signal; keeps changes small

Example UI display

AI Trust Score: 94% | Safety: High | Tests Passed

Expandable details: clone type, locations, suggested method, validation results, and rationale.

Failure-state behavior

If model / compiler / tests fail → show “Unavailable” or “Blocked” with diagnostic details instead of silently failing.

Sources: Confidence Score Brainstorm; Together AI logprobs docs; Tian et al. calibration; AlOmar et al. behavior preservation in refactoring.